

Graded All-Orders Asymptotics for Successive-Primorial Crossovers in the Odlyzko–Rubinstein–Wolf Exponential-Integral Approximation

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Abstract

Let $P = p_k^\#$ and $q = p_{k+1}$, so that $qP = p_{k+1}^\#$ is the next primorial. We study the pairwise crossover between these two candidates in the explicit Odlyzko–Rubinstein–Wolf (ORW) exponential-integral approximation

$$\mathcal{M}(X, D) = \mathfrak{S}(D)I_D(X), \quad I_D(X) = \int_2^X \frac{\exp(-D/\log t)}{(\log t)^2} dt.$$

Writing $L = \log X$, we first prove a uniform all-orders endpoint expansion for $I_D(e^L)$, with generalized Laguerre coefficients and a remainder that remains controlled under differentiation in the growing endpoint parameter. We then reorganize the logarithmic expansion into a graded asymptotic scale and rigorously invert the full ORW crossover equation. If

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}$$

denotes the exact solution of ORW’s leading exponential comparison, then for every fixed $M \geq 0$,

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(P, qP)}{L_0^m} + O_M\left(\frac{(1+qP)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}}\right),$$

with an explicit Lagrange–Bürmann coefficient generator and $\deg c_m \leq \lfloor m/2 \rfloor$. In particular,

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1+qP}{L_0^2}\right).$$

The constants 1 and 3 are universal for general endpoint pairs satisfying $D_1/L_0 \rightarrow 0$; primorial arithmetic enters only at the next polynomial coefficient. This absolute endpoint displacement is asymptotically separated from the classical algebraic expansion of L_0/P , whose first terms are $q^2 - \frac{5}{2}q + \frac{17}{12} + \dots$.

The unconditional results concern the explicitly defined ORW approximation. For actual consecutive-prime gaps we give a separate conditional transfer theorem and, through a divisor-proxy and signed Vaughan analysis, exhibit a representative short affine prime-correlation obstruction that is not covered by the currently cited results.

Keywords: jumping champions; prime gaps; primorials; singular series; generalized Laguerre polynomials; asymptotic inversion.

2020 Mathematics Subject Classification: 11N05 (primary); 11N35, 41A60 (secondary).

1 Introduction and main results

Let

$$2 = p_1 < p_2 < p_3 < \cdots$$

be the primes, and for an even integer D define

$$N(X, D) = \#\{n : p_{n+1} \leq X, p_{n+1} - p_n = D\}.$$

A value of D maximizing $N(X, D)$ is called a *jumping champion*. Erdős and Straus [2] studied the most frequent differences between consecutive primes. Odlyzko, Rubinstein, and Wolf (ORW) [4] later developed a detailed heuristic model and numerical picture in which primordial gaps control the large transitions. Goldston and Ledoan obtained rigorous consequences of Hardy–Littlewood hypotheses in this direction [5, 6, 8], and Feng and Wu treated the corresponding k -tuple problem [9].

The present paper has one unconditional analytic core: the *full crossover root of the explicit ORW exponential-integral approximation*. Put

$$P = p_k^\#, \quad q = p_{k+1}, \quad D_0 = P, \quad D_1 = qP.$$

ORW’s leading exponential comparison between the successive primordial candidates has the exact logarithmic solution

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}.$$

We take this classical leading scale as the reference point and ask for the location of the root of the *full* endpoint integral, to every fixed asymptotic order, while the endpoint parameter itself grows.

1.1 Unconditional core and headline results

The first step is a uniform endpoint theorem for

$$I_D(X) = \int_2^X \frac{\exp(-D/\log t)}{(\log t)^2} dt.$$

After the endpoint change of variables, the classical generating function for generalized Laguerre polynomials gives the coefficient structure directly. Theorem 2.1 proves, uniformly for $0 \leq D/L \leq \eta < 1$,

$$I_D(e^L) = \frac{e^{L-D/L}}{L^2} \left[\sum_{j=0}^{J-1} \frac{j! L_j^{(1)}(D/L)}{L^j} + O_{\eta, J}(L^{-J}) \right],$$

with differentiated remainder control in the endpoint parameter.

The second and principal step is the graded inversion of the crossover equation. Because $D_1 \rightarrow \infty$, ordinary fixed-parameter formal reversion is not enough. We prove degree bounds for the logarithmic coefficients, identify an asymptotic scale adapted to the growing endpoint, and only then use Lagrange–Bürmann inversion. Theorem 3.8 gives, for every fixed $M \geq 0$,

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M \left(\frac{(1 + D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}} \right),$$

where the c_m are explicit symmetric polynomials satisfying

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

The first terms are

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + \frac{12 - \frac{1}{2}(q+1)P}{L_0^2} + \frac{63 - 4(q+1)P}{L_0^3} + \cdots.$$

The general-pair extension, Proposition 3.9, shows that the first two endpoint corrections are not specifically primorial. If $D_0 < D_1$, $\mathfrak{S}(D_1) > \mathfrak{S}(D_0)$,

$$L_0 = \frac{D_1 - D_0}{\log(\mathfrak{S}(D_1)/\mathfrak{S}(D_0))}, \quad L_0 \rightarrow \infty, \quad \frac{D_1}{L_0} \rightarrow 0,$$

then Corollary 3.10 gives

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1 + D_1}{L_0^2}\right).$$

Thus the constants 1 and 3 come from the endpoint integral itself. In the successive-primorial specialization, the arithmetic scale

$$\frac{L_0}{P} = q^2 - \frac{5}{2}q + \frac{17}{12} + \cdots$$

and the absolute endpoint displacement beginning with +1 form two asymptotically separated hierarchies.

1.2 Conditional arithmetic bridge and scope

The later arithmetic sections are deliberately separated from this unconditional core. Section 4 asks what additional input is sufficient to transfer the second-order crossover coefficient from the ORW model to actual consecutive-prime gaps. Exact Bonferroni inequalities and the proved fixed-endpoint singular-series averages of Feng and Wu reduce the problem to an averaged prime-correlation hypothesis. Section 5 then diagnoses the remaining obstruction in one concrete transfer route: a normalized divisor proxy is accurate beyond the target scale, but a prime-versus-proxy remainder survives and, after signed Vaughan reassembly, leads to affine Möbius–two-prime correlations on a logarithmic shift range.

Accordingly, the paper does *not* claim an unconditional theorem about the true jumping champion. Its unconditional contribution is the Laguerre endpoint structure and graded all-orders root theory for the explicitly defined ORW approximation. The actual-prime result is conditional and pairwise, and the final section is diagnostic rather than a parity-breaking theorem.

1.3 Prior work, novelty, and priority

The Hardy–Littlewood prime-tuple conjecture [1] supplies the singular series underlying the model. Gallagher [3] connected sufficiently uniform prime-tuple conjectures with Poisson statistics in logarithmic intervals, and Montgomery and Soundararajan [10] developed refined singular-series and short-interval statistics. Modern constrained singular-series averages include work of Kuperberg [11, 12].

For the present problem, an important priority point is that precise transition locations are not themselves new. On p. 5 of their paper, ORW already write the leading exponential comparison between two successive primorial candidates; taking logarithms of that displayed comparison gives exactly the quantity L_0 used below. They then simplify it further by a first-order logarithmic approximation. ORW also explicitly use integration by parts on their heuristic integral and, for the

first fixed transitions such as $6 \rightarrow 30$ and $30 \rightarrow 210$, evaluate finite inclusion–exclusion truncations numerically using coefficients of Brent. Thus neither higher numerical precision for fixed small takeovers nor asymptotic expansion of the ORW integral in a generic sense is claimed as new here. Our claim concerns the uniform all-orders coefficient structure and rigorous graded inversion of the ORW integral root in the growing successive-primorial regime. ORW also used fixed-endpoint singular-series averages in their transition analysis, and Feng–Wu explicitly describe this role [9]. Goldston and Ledoan’s paper on consecutive gaps [6] treats a fixed-ratio regime and notes the need for sharper formulas when the normalized gap tends to zero. They also announced a sequel at the 2012 AMS Fall Eastern Sectional Meeting [7]. Feng and Wu record that, at the time of that announcement, the sharper fixed-endpoint remainder was stated without proof and prove an explicit estimate sufficient for our second-order transfer discussion [9]. Goldston and Ledoan’s later jumping-champion paper [8] also states the sharper bound in its discussion of the fixed-endpoint average. We therefore do not treat that sharper remainder as a novelty claim here.

We therefore do not claim to be the first to study precise jumping-champion transitions, nor do we claim absolute historical priority for every small-gap coefficient. Laguerre polynomials themselves also have precedent in the prime-gap literature: Goldston and Yildirim use generalized Laguerre polynomials in an orthogonal-polynomial optimization for small gaps between primes, where the smallest Laguerre zero controls the optimized moment problem [15]. That role is different from the one below, where the classical Laguerre generating function produces the endpoint coefficients of the ORW exponential integral and is then coupled to a graded root inversion. To the best of our search of the published and publicly accessible literature, we have not located an all-orders endpoint expansion of this ORW exponential integral in generalized Laguerre polynomials, nor a graded all-orders inversion of its successive-primorial crossover about the ORW leading scale. The resulting universal $1 + 3/L_0$ endpoint correction, explicit Lagrange–Bürmann coefficient generator, and graded degree hierarchy are the principal novelty claims of this paper. More specifically, the novelty claim is not the classical Laguerre generating function or formal reversion in isolation, but their uniform, differentiated use in the growing-endpoint ORW regime and the resulting rigorous root inversion. Neither L_0 itself nor the coefficient $-5/2$ in the expansion of L_0/P is claimed as an independent novelty. Because Goldston and Ledoan announced a sequel in the same small-normalized-gap regime, and later related work contains sharper singular-series information, we deliberately avoid an absolute first-priority claim beyond the published and publicly accessible formulas that we have been able to compare directly. Repeated asymptotic expansion, Laguerre polynomials, formal inversion, singular-series averaging, sieve methods, and truncated divisor sums are classical tools and are not claimed as new in isolation.

The unconditional results in Sections 2–3 are statements about the explicitly defined approximation $\mathcal{M}$, not unconditional theorems about the true jumping champion. The arithmetic transfer theorem in Section 4 is conditional and pairwise; it does not exclude all competing gaps.

2 Uniform Laguerre endpoint expansion for the ORW integral

For even D , write

$$\mathfrak{S}(D) = 2C_2 \prod_{\substack{p|D \\ p>2}} \frac{p-1}{p-2}.$$

For successive primorials,

$$\frac{\mathfrak{S}(qP)}{\mathfrak{S}(P)} = \frac{q-1}{q-2}. \quad (2.1)$$

Define

$$\mathcal{M}(X, D) = \mathfrak{S}(D)I_D(X), \quad I_D(X) = \int_2^X \frac{e^{-D/\log t}}{(\log t)^2} dt,$$

and set $L = \log X$. The prime number theorem in Chebyshev form gives

$$\log P = \vartheta(p_k) \sim p_k \sim q, \quad P = e^{(1+o(1))q}. \quad (2.2)$$

The transition scale obtained below satisfies

$$L \asymp Pq^2, \quad q \sim \log L \sim \log \log X. \quad (2.3)$$

In particular, for every fixed $A > 0$,

$$L^{-1} = o(q^{-A}). \quad (2.4)$$

2.1 A generating-function proof and differentiated remainder

The generalized Laguerre structure in the endpoint expansion is not merely an algebraic consequence of repeated differentiation. It is already present in the integral through the classical Laguerre generating function.

Put

$$z = \frac{1}{L}, \quad x = \frac{D}{L}.$$

With $u = L - s$, one obtains

$$I_D(e^L) = \frac{e^L e^{-x}}{L^2} \mathcal{J}(z, x),$$

where

$$\mathcal{J}(z, x) = \int_0^{L-\log 2} e^{-s} (1-sz)^{-2} \exp\left(-x\left((1-sz)^{-1} - 1\right)\right) ds.$$

Since

$$(1-sz)^{-1} - 1 = \frac{sz}{1-sz},$$

the classical generating function

$$\sum_{j \geq 0} L_j^{(1)}(x) t^j = (1-t)^{-2} \exp\left(-\frac{xt}{1-t}\right)$$

gives, formally with $t = sz$,

$$(1-sz)^{-2} \exp\left(-x \frac{sz}{1-sz}\right) = \sum_{j \geq 0} L_j^{(1)}(x) (sz)^j.$$

Theorem 2.1 (All-orders endpoint expansion and differentiated remainder). *Fix $J \geq 1$, $r \geq 0$, and $0 < \eta < 1$. Uniformly for $0 \leq x \leq \eta$ as $z \rightarrow 0^+$,*

$$\frac{\partial^r}{\partial x^r} \left[\mathcal{J}(z, x) - \sum_{j=0}^{J-1} j! L_j^{(1)}(x) z^j \right] = O_{\eta, J, r}(z^J).$$

Consequently,

$$I_D(e^L) = \frac{e^L e^{-D/L}}{L^2} \left[\sum_{j=0}^{J-1} \frac{j! L_j^{(1)}(D/L)}{L^j} + O_{\eta, J}(L^{-J}) \right],$$

and the normalized remainder may be differentiated with respect to D , with each D -derivative gaining one further factor L^{-1} .

Proof. Split the s -integral at

$$S = z^{-1/2}.$$

On $0 \leq s \leq S$, one has

$$|sz| \leq z^{1/2},$$

so for sufficiently small z the Laguerre generating function is analytic in a fixed disk containing every $t = sz$ under consideration. Taylor's theorem, uniformly for $x \in [0, \eta]$ and after any fixed number r of x -derivatives, gives

$$\frac{\partial^r}{\partial x^r} \left[(1 - sz)^{-2} \exp \left(-x \frac{sz}{1 - sz} \right) - \sum_{j=0}^{J-1} L_j^{(1)}(x) (sz)^j \right] = O_{\eta, J, r}((sz)^J).$$

Multiplying by e^{-s} and integrating yields

$$O_{\eta, J, r} \left(z^J \int_0^\infty e^{-s} s^J ds \right) = O_{\eta, J, r}(z^J).$$

On the complementary interval

$$S \leq s \leq z^{-1} - \log 2,$$

one has

$$1 - sz \geq z \log 2.$$

The exponential factor involving x has absolute value at most one for $x \geq 0$, while an x -derivative of order r introduces at most the factor

$$\left(\frac{sz}{1 - sz} \right)^r = O_r(z^{-r}).$$

Together with $(1 - sz)^{-2} = O(z^{-2})$, the tail is therefore bounded by

$$O_r \left(z^{-r-2} \int_{z^{-1/2}}^\infty e^{-s} ds \right) = O_r \left(z^{-r-2} e^{-z^{-1/2}} \right) = O_{r, A}(z^A)$$

for every fixed $A > 0$. The same estimate applies to each term of the finite Laguerre truncation, so the complementary interval is negligible compared with z^J .

Finally,

$$\int_0^\infty e^{-s} s^j ds = \Gamma(j+1) = j!,$$

which gives the claimed coefficients.

Since $x = Dz$,

$$\frac{\partial}{\partial D} = z \frac{\partial}{\partial x}.$$

Thus every D -derivative of the normalized remainder gains one additional factor $z = L^{-1}$. $\square$

Remark 2.2. *Theorem 2.1 supplies the uniform differentiated remainder required in the proof of the graded all-orders root theorem. In particular, when the normalized endpoint remainders at D_1 and D_0 are subtracted and divided by $A = D_1 - D_0$, the mean value theorem contributes one D -derivative and hence one additional factor L^{-1} . This prevents the endpoint remainder from contaminating the coefficient scale in the Lagrange inversion.*

Remark 2.3 (Role of the Laguerre representation). *The occurrence of Laguerre polynomials in Theorem 2.1 is a compact organization of an endpoint asymptotic expansion, rather than a claim that Laguerre asymptotics themselves are new. The point needed later is that the generating-function representation gives all coefficients together with differentiated uniform remainders, which remain usable when the endpoint parameter D grows with the successive primorials.*

2.2 Monotonicity, uniqueness, and localization of the crossover

Let

$$w(t) = \frac{e^{-P/\log t}}{(\log t)^2}, \quad r(t) = e^{-(q-1)P/\log t}.$$

Then

$$\frac{I_{qP}(X)}{I_P(X)} = \frac{\int_2^X w(t)r(t) dt}{\int_2^X w(t) dt}.$$

The function $r(t)$ is strictly increasing. Hence the ratio above is a weighted average strictly below $r(X)$, and differentiation gives the following.

Lemma 2.4 (Monotonicity). *The map*

$$X \mapsto \frac{I_{qP}(X)}{I_P(X)}$$

is strictly increasing. Therefore the model crossover, if it exists, is unique.

Put

$$A = (q-1)P, \quad s = \log \frac{q-1}{q-2}, \quad L_0 = \frac{A}{s}, \quad S = (q+1)P. \quad (2.5)$$

From the logarithmic endpoint coefficients derived below,

$$\begin{aligned} F(L) &:= \log \frac{\mathcal{M}(e^L, qP)}{\mathcal{M}(e^L, P)} \\ &= s - \frac{A}{L} - \frac{A}{L^2} - \frac{4A}{L^3} + \frac{A(S/2 - 22)}{L^4} + \frac{A(6S - 148)}{L^5} + O\left(\frac{A(1 + S^2)}{L^6}\right). \end{aligned} \quad (2.6)$$

Since $s = A/L_0$, this gives

$$F(L_0) = -\frac{A}{L_0^2}(1 + o(1)) < 0$$

and

$$F(L_0 + 2) = \frac{A}{L_0^2}(1 + o(1)) > 0.$$

Consequently, for sufficiently large q ,

$$L_0 < L_{\text{int}} < L_0 + 2. \quad (2.7)$$

This a priori localization is used before the graded root inversion and prevents circularity.

3 Graded inversion of the full ORW crossover root

3.1 Graded coefficients and rigorous root inversion

Put

$$z = \frac{1}{L}, \quad D_0 = P, \quad D_1 = qP, \quad A = D_1 - D_0 = (q-1)P,$$

and

$$t = \frac{1}{L_0} = \frac{1}{A} \log \frac{q-1}{q-2}.$$

For the root analysis, reparameterize the normalized endpoint integral by

$$\mathcal{J}_D(z) = \frac{I_D(e^{1/z})}{e^{1/z} e^{-Dz} z^2}.$$

Equivalently,

$$\mathcal{J}_D(z) = \int_0^{z^{-1} - \log 2} e^{-s} (1 - sz)^{-2} \exp\left(-Dz \frac{sz}{1 - sz}\right) ds.$$

By Theorem 2.1, for every fixed J ,

$$\mathcal{J}_D(z) = \sum_{j=0}^{J-1} j! L_j^{(1)}(Dz) z^j + O_J(z^J)$$

uniformly whenever $0 \leq Dz \leq \eta < 1$. The same remainder may be differentiated with respect to D , with one additional factor z .

Define the formal series

$$\mathcal{B}_D(z) = \sum_{j \geq 0} j! L_j^{(1)}(Dz) z^j = 1 + \sum_{n \geq 1} b_n(D) z^n$$

and

$$\mathcal{H}_D(z) = \log \mathcal{B}_D(z) = \sum_{n \geq 1} h_n(D) z^n.$$

Lemma 3.1 (Explicit endpoint coefficients). *For every $n \geq 1$,*

$$b_n(D) = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m \frac{(n-m)!}{m!} \binom{n-m+1}{m+1} D^m.$$

Hence

$$\deg_D b_n \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Use

$$L_j^{(1)}(x) = \sum_{m=0}^j (-1)^m \binom{j+1}{m+1} \frac{x^m}{m!}.$$

The contribution indexed by (j, m) is proportional to $D^m z^{j+m}$. Setting $j = n - m$ gives the formula, and $m \leq j$ is equivalent to $m \leq n/2$. $\square$

Lemma 3.2 (Logarithmic coefficient recurrence). *Let $b_0 = 1$. Then*

$$h_n = b_n - \frac{1}{n} \sum_{k=1}^{n-1} k h_k b_{n-k}, \quad n \geq 1,$$

and

$$\deg_D h_n \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. From

$$\mathcal{B}'_D(z) = \mathcal{H}'_D(z)\mathcal{B}_D(z)$$

compare the coefficient of z^{n-1} .

For the degree bound, every contribution to h_n arising from $\log(1+U)$ is a product

$$b_{n_1} \cdots b_{n_r}, \quad n_1 + \cdots + n_r = n.$$

Its degree is at most

$$\sum_i \left\lfloor \frac{n_i}{2} \right\rfloor \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

□

The first coefficients are

$$h_1(D) = 2,$$

$$h_2(D) = 4 - D,$$

$$h_3(D) = \frac{44}{3} - 4D,$$

$$h_4(D) = 74 - 22D + \frac{1}{2}D^2,$$

$$h_5(D) = \frac{2312}{5} - 148D + 6D^2,$$

and

$$h_6(D) = \frac{10192}{3} - 1156D + 64D^2 - \frac{1}{3}D^3.$$

Lemma 3.3 (Graded logarithmic remainder). *Fix $N \geq 1$ and $0 < \eta < 1$. Let*

$$r_N = \left\lfloor \frac{N+1}{2} \right\rfloor.$$

Uniformly in the regime

$$0 < z \rightarrow 0, \quad 0 \leq Dz \leq \eta,$$

one has

$$\log \mathcal{J}_D(z) = \sum_{n=1}^N h_n(D)z^n + O_{N,\eta}\left((1+D)^{r_N}z^{N+1}\right).$$

Moreover,

$$\frac{\partial}{\partial D} \left[\log \mathcal{J}_D(z) - \sum_{n=1}^N h_n(D)z^n \right] = O_{N,\eta}\left((1+D)^{r_N-1}z^{N+1}\right),$$

with the convention that $(1+D)^{-1}$ is replaced by 1 if $r_N = 0$.

Proof. Apply the uniform Laguerre endpoint expansion with $J = N+1$:

$$\mathcal{J}_D(z) = \sum_{j=0}^N j! L_j^{(1)}(Dz)z^j + O_{N,\eta}(z^{N+1}).$$

Expanding the finite Laguerre polynomials gives

$$\sum_{j=0}^N j! L_j^{(1)}(Dz)z^j = 1 + \sum_{n=1}^{2N} \tilde{b}_{n,N}(D)z^n,$$

where

$$\tilde{b}_{n,N} = b_n \quad (n \leq N)$$

and, uniformly in $n \leq 2N$,

$$\deg_D \tilde{b}_{n,N} \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

For $n \geq N + 1$, define the model size

$$R_n = (1 + D)^{\lfloor n/2 \rfloor} z^n.$$

Successive ratios satisfy

$$\frac{R_{n+1}}{R_n} = \begin{cases} z, & n \text{ even}, \\ (1 + D)z, & n \text{ odd}. \end{cases}$$

Under the stated uniform hypothesis $Dz \leq \eta < 1$, one has

$$(1 + D)z = Dz + z \leq \eta + z.$$

Hence, for sufficiently small z , this ratio is bounded by a fixed constant strictly below one (for example $(1 + \eta)/2$). Thus, for fixed N , the omitted finite polynomial terms are dominated by the first omitted graded scale:

$$\sum_{n=N+1}^{2N} \tilde{b}_{n,N}(D)z^n = O_{N,\eta}\left((1 + D)^{r_N} z^{N+1}\right).$$

The original endpoint remainder $O(z^{N+1})$ is smaller than this bound.

The truncated expression differs from 1 by $O_{N,\eta}(z)$ uniformly for $Dz \leq \eta < 1$. Indeed, every monomial $D^m z^n$ occurring in a coefficient with $m \leq \lfloor n/2 \rfloor$ may be written as $(Dz)^m z^{n-m}$, and $n - m \geq 1$. Therefore the logarithm is analytic in a fixed neighborhood of 1, and formal coefficient comparison gives

$$\log \mathcal{J}_D(z) = \sum_{n=1}^N h_n(D)z^n + O_{N,\eta}\left((1 + D)^{r_N} z^{N+1}\right).$$

For the differentiated estimate, differentiate the finite polynomial term by term. Differentiation lowers the degree in D by one. The differentiated endpoint remainder gains an extra factor z from

$$\partial_D = z \partial_x, \quad x = Dz,$$

and is therefore smaller than the claimed graded bound. Differentiating the logarithm introduces only uniformly bounded factors because $\mathcal{J}_D(z) = 1 + O(z)$. $\square$

For $n \geq 2$, define

$$a_n(D_0, D_1) = \frac{h_n(D_1) - h_n(D_0)}{D_1 - D_0}.$$

Then a_n is symmetric and

$$\deg a_n \leq \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

Define

$$\Phi(z) = z - \sum_{n \geq 2} a_n z^n.$$

For the successive primorial pair,

$$\begin{aligned}\Phi(z) &= z + z^2 + 4z^3 + \left(22 - \frac{S}{2}\right)z^4 + (148 - 6S)z^5 \\ &\quad + \left(1156 - 64S + \frac{Q}{3}\right)z^6 + \cdots,\end{aligned}$$

where

$$S = D_0 + D_1$$

and

$$Q = D_0^2 + D_0D_1 + D_1^2.$$

Since $\Phi(z) = z + O(z^2)$, it possesses a unique formal inverse (see, e.g., [20])

$$\Psi(t) = \Phi^{\langle -1 \rangle}(t) = \sum_{n \geq 1} \psi_n t^n.$$

Lemma 3.4 (Inverse-coefficient degree bound). *For $n \geq 2$,*

$$\deg \psi_n \leq \left\lfloor \frac{n-2}{2} \right\rfloor.$$

Proof. Put

$$G(u) = \frac{\Phi(u)}{u} = 1 + \sum_{r \geq 1} g_r u^r.$$

Then

$$g_r = -a_{r+1}, \quad \deg g_r \leq \left\lfloor \frac{r-1}{2} \right\rfloor.$$

Lagrange inversion gives

$$\psi_n = \frac{1}{n} [u^{n-1}] G(u)^{-n}.$$

For $n \geq 2$, every nonconstant monomial contributing to the coefficient of u^{n-1} contains factors $g_{r_1}, \dots, g_{r_\ell}$ with

$$r_1 + \cdots + r_\ell = n - 1.$$

Hence its degree is at most

$$\sum_i \left\lfloor \frac{r_i - 1}{2} \right\rfloor \leq \left\lfloor \frac{n-2}{2} \right\rfloor.$$

□

Lemma 3.5 (Graded truncation of the formal inverse). *Let*

$$\Phi_K(z) = z - \sum_{n=2}^K a_n z^n, \quad Z_K(t) = \sum_{n=1}^K \psi_n t^n,$$

where the ψ_n are the coefficients of the formal inverse of Φ . Then, for every fixed $K \geq 2$,

$$t - \Phi_K(Z_K(t)) = O_K\left((1 + D_1)^{\lfloor (K-1)/2 \rfloor} t^{K+1}\right)$$

in the successive-primorial regime $D_1 t \rightarrow 0$.

Proof. Formal inversion gives cancellation of every coefficient of t^r for $2 \leq r \leq K$. Assign to a coefficient multiplying t^r the polynomial degree bound

$$\deg_D \leq \left\lfloor \frac{r-2}{2} \right\rfloor.$$

To see explicitly that the bound is preserved under composition, consider a monomial from $a_j Z_K(t)^j$, with indices $n_1 + \dots + n_j = r$. We have

$$2 \deg a_j \leq j - 2,$$

and, if s of the indices n_i are at least 2,

$$2 \sum_i \deg \psi_{n_i} \leq \sum_{n_i \geq 2} (n_i - 2).$$

Therefore

$$2 \deg \left(a_j \prod_i \psi_{n_i} \right) \leq j - 2 + \sum_{n_i \geq 2} (n_i - 2) \leq r - 2.$$

Thus every coefficient multiplying t^r in the finite composition has degree at most $\lfloor (r-2)/2 \rfloor$. Hence the coefficient of t^r , $r \geq K+1$, in the fixed truncation has size

$$O_K \left((1 + D_1)^{\lfloor (r-2)/2 \rfloor} \right).$$

For the corresponding graded terms

$$U_r = (1 + D_1)^{\lfloor (r-2)/2 \rfloor} t^r$$

the successive ratios are either t or $(1 + D_1)t$. Both tend to zero in the successive-primorial regime. Since only finitely many terms arise from the fixed polynomial truncation, they are dominated by the first uncanceled scale $r = K+1$, which proves the claim. $\square$

Definition 3.6 (Graded root scale). *For $m \geq 0$, define*

$$E_m(q) = \frac{(1 + D_1)^{\lfloor m/2 \rfloor}}{L_0^m}.$$

We say that a family $R(q)$ has a graded generalized asymptotic expansion

$$R(q) = \sum_{m \geq 0}^{\text{gr}} R_m(q)$$

with respect to $\{E_m\}$ if, for every fixed M ,

$$R(q) - \sum_{m=0}^M R_m(q) = O_M(E_{M+1}(q)), \quad R_m(q) = O_m(E_m(q)).$$

We use this terminology in the explicit sense defined here: the successive envelopes form an asymptotic scale, and each fixed truncation has a remainder bounded by the next envelope. This is modeled on the standard use of asymptotic scales; see, for example, Olver [23].

Lemma 3.7 (The graded envelopes form an asymptotic scale). *In the successive-primorial regime,*

$$E_{m+1} = o(E_m) \quad (m \geq 0 \text{ fixed}).$$

More precisely,

$$\frac{E_{2r+1}}{E_{2r}} = \frac{1}{L_0},$$

whereas

$$\frac{E_{2r+2}}{E_{2r+1}} = \frac{1 + D_1}{L_0} \sim \frac{1}{q}.$$

Proof. The first identity is immediate. For the second, use

$$D_1 = qP, \quad L_0 \sim Pq^2.$$

Thus both ratios tend to zero. □

It is useful to make the two scales dimensionless. Put

$$\alpha = \frac{D_1}{L_0} \asymp \frac{1}{q}, \quad \beta = \frac{1}{L_0}.$$

A monomial D_1^d/L_0^m can then be written as

$$\alpha^d \beta^{m-d}.$$

The degree restriction $d \leq \lfloor m/2 \rfloor$ confines the root corrections to the cone $m - d \geq d$. The dominant envelopes are therefore interlaced as

$$1, \quad \beta, \quad \alpha\beta, \quad \alpha\beta^2, \quad \alpha^2\beta^2, \quad \alpha^2\beta^3, \quad \dots$$

This is the precise sense in which the root expansion is a two-scale graded hierarchy.

Theorem 3.8 (All-orders graded successive-primorial root expansion). *For every fixed integer $M \geq 0$, there exist symmetric polynomials $c_m(D_0, D_1)$, $0 \leq m \leq M$, such that*

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M \left(\frac{(1 + D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}} \right).$$

Moreover,

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor,$$

and therefore

$$\frac{c_m(D_0, D_1)}{L_0^m} = O_m(E_m(q)).$$

Consequently the displayed formula is a graded generalized asymptotic expansion in the sense of Definition 3.6.

If

$$G(u) = \frac{\Phi(u)}{u},$$

then the coefficients are generated by

$$c_m = \frac{1}{m+1} [u^m] G'(u) \left(\frac{u}{\Phi(u)} \right)^{m+1}.$$

At the formal level, the coefficient generator is equivalently encoded by

$$z_{\text{formal}} = \Psi(t), \quad t = \frac{1}{L_0},$$

followed by formal reciprocal expansion. No convergence of Ψ at $t = 1/L_0$ is asserted or required.

Proof. Let

$$F(L) = \log \frac{\mathfrak{S}(D_1)I_{D_1}(e^L)}{\mathfrak{S}(D_0)I_{D_0}(e^L)}.$$

For successive primorials,

$$\frac{1}{A} \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)} = t.$$

Apply Lemma 3.3 and the mean value theorem in the parameter D . For every fixed $N \geq 2$,

$$\frac{F(1/z)}{A} = t - z + \sum_{n=2}^N a_n z^n + O_N\left((1 + D_1)^{\lfloor (N+1)/2 \rfloor - 1} z^{N+1}\right)$$

uniformly in the root neighborhood. Here

$$D_1 z \asymp q^{-1} \rightarrow 0.$$

Set $N = M + 2$, and let

$$z_M(t) = \sum_{n=1}^{M+2} \psi_n t^n.$$

Apply Lemma 3.5 with $K = M + 2$. Since $D_1 t \asymp q^{-1} \rightarrow 0$, it gives

$$t - \Phi_{M+2}(z_M) = O_M\left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3}\right).$$

The truncated analytic crossover polynomial is exactly $t - \Phi_{M+2}(z)$. Therefore the inverse-truncation estimate may be combined directly with the analytic remainder in the preceding expansion, giving

$$\frac{F(1/z_M)}{A} = O_M\left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3}\right).$$

The previously established localization

$$L_0 < L_{\text{int}} < L_0 + 2$$

implies

$$z_{\text{int}} = t + O(t^2).$$

To control the derivative without differentiating an asymptotic remainder in z , use the exact endpoint identity

$$\frac{d}{dL} I_D(e^L) = \frac{e^{L-D/L}}{L^2}, \quad \frac{d}{dL} \log I_D(e^L) = \frac{1}{\mathcal{J}_D(1/L)}.$$

Therefore

$$\frac{d}{dz} \left(\frac{F(1/z)}{A} \right) = -\frac{z^{-2}}{A} \left(\frac{1}{\mathcal{J}_{D_1}(z)} - \frac{1}{\mathcal{J}_{D_0}(z)} \right).$$

Theorem 2.1 with $J = 2$, together with its D -differentiated remainder, gives uniformly in the root neighborhood

$$\frac{\mathcal{J}_{D_1}(z) - \mathcal{J}_{D_0}(z)}{A} = -z^2 + O(z^3), \quad \mathcal{J}_{D_i}(z) = 1 + O(z).$$

Consequently

$$\frac{d}{dz} \left(\frac{F(1/z)}{A} \right) = -1 + O(z) = -1 + o(1)$$

uniformly in this neighborhood. Hence the derivative is bounded away from zero for sufficiently large q . By the mean value theorem,

$$z_{\text{int}} - z_M = O_M \left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3} \right).$$

Since both z_{int} and z_M are asymptotic to t , taking reciprocals yields

$$L_{\text{int}} = \frac{1}{z_M} + O_M \left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+1} \right).$$

To identify the coefficients, use

$$t = \Phi(\Psi(t))$$

and therefore

$$\frac{t}{\Psi(t)} = G(\Psi(t)).$$

At the level of formal coefficient extraction,

$$\frac{1}{\Psi(t)} = \frac{1}{t} G(\Psi(t))$$

as an identity of formal Laurent series. Since the analytic root has already been matched to every fixed truncation above, this identity is used only to identify its coefficients. Lagrange–Bürmann gives

$$[t^n] G(\Psi(t)) = \frac{1}{n} [u^{n-1}] G'(u) \left(\frac{u}{\Phi(u)} \right)^n,$$

which yields the displayed formula for c_m .

Finally, write

$$G(u) = 1 + \sum_{j \geq 1} g_j u^j, \quad \deg g_j \leq \left\lfloor \frac{j-1}{2} \right\rfloor \leq \left\lfloor \frac{j}{2} \right\rfloor.$$

Hence the coefficient of u^r in $G'(u)$ has degree at most $\lfloor r/2 \rfloor$. Every monomial contributing to the coefficient of u^s in the fixed power $G(u)^{-(m+1)}$ is a product $g_{j_1} \cdots g_{j_\ell}$ with $j_1 + \cdots + j_\ell = s$, and therefore has degree at most

$$\sum_i \left\lfloor \frac{j_i}{2} \right\rfloor \leq \left\lfloor \frac{s}{2} \right\rfloor.$$

For $r + s = m$, the product appearing in $[u^m]$ consequently has degree at most

$$\left\lfloor \frac{r}{2} \right\rfloor + \left\lfloor \frac{s}{2} \right\rfloor \leq \left\lfloor \frac{m}{2} \right\rfloor,$$

which proves

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

□

3.2 General endpoint pairs and universal corrections

The coefficient system produced above is not intrinsically tied to primordial arithmetic. The next proposition isolates the endpoint conditions under which the same graded inversion applies.

Proposition 3.9 (General pair extension). *Let $D_0 = D_0(\nu) < D_1 = D_1(\nu)$ be a family of even gaps such that*

$$\mathfrak{S}(D_1) > \mathfrak{S}(D_0).$$

Put

$$A = D_1 - D_0, \quad s = \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)}, \quad L_0 = \frac{A}{s}.$$

Assume

$$L_0 \rightarrow \infty, \quad \frac{D_1}{L_0} \rightarrow 0.$$

Then, for sufficiently large ν , the model equation

$$\mathfrak{S}(D_1)I_{D_1}(e^L) = \mathfrak{S}(D_0)I_{D_0}(e^L)$$

has a unique root L_{int} in

$$L_0 < L_{\text{int}} < L_0 + 2,$$

and, for every fixed $M \geq 0$,

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M \left(\frac{(1 + D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}} \right),$$

with the same symmetric coefficient polynomials as in Theorem 3.8. Moreover the envelopes

$$E_m^{\text{gen}} = \frac{(1 + D_1)^{\lfloor m/2 \rfloor}}{L_0^m}$$

form an asymptotic scale under the present assumptions.

Thus the formal coefficient system is a property of the endpoint integral itself; successive primorials enter by supplying the particular singular-series ratio and the more specific relation $D_1/L_0 \asymp q^{-1}$.

Proof. The monotonicity argument of Section 2 uses only $D_1 > D_0$. From the logarithmic endpoint expansion and the mean value theorem in D , uniformly for $L \asymp L_0$,

$$F(L) = s - \frac{A}{L} - \frac{A}{L^2} + O \left(\frac{A}{L^3} + \frac{AD_1}{L^4} \right).$$

Since $s = A/L_0$, at the left endpoint

$$F(L_0) = -\frac{A}{L_0^2} + o \left(\frac{A}{L_0^2} \right) < 0.$$

At the right endpoint,

$$s - \frac{A}{L_0 + 2} = \frac{2A}{L_0(L_0 + 2)},$$

and therefore

$$F(L_0 + 2) = \frac{A}{L_0^2}(1 + o(1)) > 0.$$

Indeed, the remaining terms are $o(A/L_0^2)$ because $L_0 \rightarrow \infty$ and $D_1/L_0 \rightarrow 0$. Existence follows by continuity and uniqueness from monotonicity, giving the asserted localization. In the proof of Theorem 3.8, replace the primordial identity

$$\frac{1}{A} \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)} = t$$

by the definition $t = s/A = 1/L_0$. Every subsequent estimate uses only $t \rightarrow 0$ and $D_1 t \rightarrow 0$, which are precisely the present assumptions. Finally,

$$\frac{E_{2r+1}^{\text{gen}}}{E_{2r}^{\text{gen}}} = L_0^{-1} \rightarrow 0, \quad \frac{E_{2r+2}^{\text{gen}}}{E_{2r+1}^{\text{gen}}} = \frac{1 + D_1}{L_0} \rightarrow 0,$$

so the displayed envelopes indeed form an asymptotic scale. $\square$

Corollary 3.10 (Universal first endpoint corrections). *Under the hypotheses of Proposition 3.9,*

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1 + D_1}{L_0^2}\right).$$

The constants 1 and 3 are independent of the arithmetic structure of D_0, D_1 .

Proof. The logarithmic endpoint coefficients satisfy

$$h_2(D) = 4 - D, \quad h_3(D) = \frac{44}{3} - 4D.$$

Hence their divided differences are universally

$$a_2 = -1, \quad a_3 = -4.$$

Therefore

$$\Phi(z) = z + z^2 + 4z^3 + O((1 + D_1)z^4),$$

and Lagrange–Bürmann inversion gives

$$c_0 = 1, \quad c_1 = 3.$$

The $M = 1$ remainder in the graded theorem is

$$O\left(\frac{1 + D_1}{L_0^2}\right).$$

$\square$

Corollary 3.11 (First coefficients). *With*

$$S = D_0 + D_1, \quad Q = D_0^2 + D_0 D_1 + D_1^2,$$

the first coefficients are

$$c_0 = 1, \quad c_1 = 3,$$

$$c_2 = 12 - \frac{S}{2},$$

$$c_3 = 63 - 4S,$$

$$c_4 = 420 - \frac{63}{2}S + \frac{Q}{3},$$

$$c_5 = 3390 - 280S - \frac{3}{4}S^2 + 6Q,$$

and

$$c_6 = \frac{1}{4} \left(S^3 - 2SQ + 336Q - 56S^2 - 11300S + 127008 \right).$$

Remark 3.12 (Why “graded all-orders” is the precise terminology). *The coefficients $c_m(D_0, D_1)$ vary with the asymptotic parameter, so the theorem is not being presented as a one-parameter Poincaré series with fixed numerical coefficients. Instead, the explicit envelopes E_m of Definition 3.6 form an asymptotic scale by Lemma 3.7, and the remainder after every fixed truncation is bounded by the next envelope. This is a generalized asymptotic expansion to all fixed orders. No convergence of the infinite formal inverse is asserted.*

3.3 Successive-primorial specialization and the two-scale hierarchy

The exact identity

$$s = \log \frac{q-1}{q-2} = \sum_{n \geq 1} \frac{2^n - 1}{nq^n} \quad (3.1)$$

yields, by ordinary series inversion,

$$\begin{aligned} \frac{L_0}{P} &= q^2 - \frac{5}{2}q + \frac{17}{12} - \frac{1}{24q} - \frac{49}{720q^2} - \frac{163}{1440q^3} \\ &\quad - \frac{11573}{60480q^4} + O(q^{-5}). \end{aligned} \quad (3.2)$$

Therefore

$$L_{\text{int}} = Pq^2 - \frac{5}{2}Pq + O(P). \quad (3.3)$$

The coefficient $-5/2$ can also be read directly at the transition scale. Write

$$L = Pq^2 \left(1 + \frac{\theta}{q} \right), \quad |\theta| = O(1).$$

Then

$$\log \frac{q-1}{q-2} = \frac{1}{q} + \frac{3}{2q^2} + O(q^{-3}),$$

while

$$\frac{(q-1)P}{L} = \frac{1}{q} - \frac{\theta+1}{q^2} + O(q^{-3}).$$

All endpoint-integral corrections are smaller than every fixed power of q^{-1} at this scale, by (2.4), and hence

$$F(L) = \frac{\theta + 5/2}{q^2} + O(q^{-3}). \quad (3.4)$$

Separation of scales. Define

$$E(q) = \frac{q-1}{\log((q-1)/(q-2))}.$$

Then $L_0 = PE(q)$, while the root theorem gives

$$L_{\text{int}} = PE(q) + 1 + O((Pq^2)^{-1}). \quad (3.5)$$

Since $P = e^{(1+o(1))q}$,

$$\frac{L_{\text{int}}}{P} = E(q) + e^{-(1+o(1))q}. \quad (3.6)$$

Thus the corrections in (3.2) live at algebraic orders in q^{-1} after normalization by P , whereas the absolute +1 displacement generated by the integral is beyond every fixed algebraic order on that normalized scale.

3.4 Relation to ORW and fixed-order precision

The leading transition comparison already appears explicitly in ORW. In their half-gap notation, the displayed exponential comparison on p. 5 becomes, in the present actual-gap notation,

$$\frac{q-1}{q-2} \exp\left(-\frac{D_1}{L}\right) > \exp\left(-\frac{D_0}{L}\right). \quad (3.7)$$

Taking logarithms shows that its exact equality point is

$$L = L_0 = \frac{D_1 - D_0}{\log((q-1)/(q-2))} = \frac{P(q-1)}{\log((q-1)/(q-2))}. \quad (3.8)$$

Thus L_0 is best viewed as the exact ORW *leading endpoint crossover scale*, not as a new transition formula introduced here.

ORW then replace

$$\log \frac{q-1}{q-2} = \log \left(1 + \frac{1}{q-2}\right)$$

by its first-order approximation $1/(q-2)$, obtaining the simpler rough rule

$$L_{\text{poly}} = P(q-1)(q-2) = P(q^2 - 3q + 2). \quad (3.9)$$

Expanding the exact ORW leading scale instead gives

$$\frac{L_0}{P} = q^2 - \frac{5}{2}q + \frac{17}{12} + O(q^{-1}). \quad (3.10)$$

Accordingly, the coefficient $-5/2$ is an explicit algebraic coefficient of the exact ORW leading scale; it is not claimed here as a separate novelty.

The new asymptotic object studied in this paper is the difference between the exact root of the full exponential integral and this leading ORW scale:

$$L_{\text{int}} - L_0 = 1 + \frac{3}{L_0} + \frac{12 - S/2}{L_0^2} + \dots \quad (3.11)$$

Thus the refinement is not the introduction of L_0 , but the graded all-orders resolution of the full integral crossover around that already implicit leading scale.

3.5 Numerical consistency check

For numerical evaluation it is convenient to use

$$\frac{I_D(e^L)}{e^L} = \int_0^{L-\log 2} e^{-s} \frac{e^{-D/(L-s)}}{(L-s)^2} ds.$$

The table compares the numerically computed displacement $\Delta = L_{\text{int}} - L_0$ with successive truncations of the root expansion. The numerical calculation is not used in any proof.

q	Numerical Δ	$1 + 3/L_0$	through L_0^{-2}	through L_0^{-4}
5	1.0349513526	1.0359602591	1.0350981656	1.0349516001
7	1.0029269673	1.0030386926	1.0029278888	1.0029269671
11	1.0001473723	1.0001505150	1.0001473736	1.0001473723
13	1.0000092576	1.0000094168	1.0000092576	1.0000092576
17	1.0000003981	1.0000004030	1.0000003981	1.0000003981

Reproducibility. The numerical values in the table were computed independently of the asymptotic coefficient algebra. We evaluated the normalized endpoint integral after the change of variables

$$\frac{I_D(e^L)}{e^L} = \int_0^{L-\log 2} e^{-s} \frac{e^{-D/(L-s)}}{(L-s)^2} ds$$

using 80-decimal-digit arithmetic and adaptive quadrature, and solved $F(L) = 0$ in the already proved interval $(L_0, L_0 + 2)$. The reported digits are stable under increasing the working precision and quadrature cutoff. A short ancillary script, `numerical_check.py`, reproduces the table and is supplied with the source files. These computations are consistency checks only and are not used in any proof.

4 Conditional bridge to actual consecutive-prime gaps

Status of this section. Everything in this section concerns a *conditional transfer beyond the unconditional ORW model theorem*. No statement below upgrades Theorems 2.1 or 3.8 to an unconditional result about actual consecutive primes. The purpose is to identify a sufficient second-order arithmetic input as sharply as possible.

Define

$$Y_D(n) = \mathbf{1}_{\mathbb{P}}(n) \mathbf{1}_{\mathbb{P}}(n+D), \quad K_D(n) = \#\{p : n < p < n+D\}.$$

Then

$$N(X, D) = \sum_{n+D \leq X} Y_D(n) \mathbf{1}_{\{K_D(n)=0\}}. \quad (4.1)$$

For $j \geq 2$, set

$$T_j(X, D) = \sum_{n+D \leq X} Y_D(n) \binom{K_D(n)}{j-2}, \quad B_D = T_2(X, D). \quad (4.2)$$

Pointwise Bonferroni inequalities give the exact finite bounds

$$B_D - T_3 + T_4 - T_5 \leq N(X, D) \leq B_D - T_3 + T_4. \quad (4.3)$$

Combining the fixed-dimensional upper-bound sieve (see, e.g., [17]) with Lemma 4.1 for $r = 5$ gives, for $D = O(L)$,

$$\begin{aligned} T_5(X, D) &\ll \frac{X}{L^5} \sum_{0 < h_1 < h_2 < h_3 < D} \mathfrak{S}(\{0, h_1, h_2, h_3, D\}) + O(D^3) \\ &\ll \mathfrak{S}(D) \frac{X}{L^2} \left(\frac{D}{L}\right)^3. \end{aligned} \tag{4.4}$$

Here the harmless $O(D^3)$ term covers inadmissible tuples for which the singular-series upper-bound formulation is vacuous; it is negligible on the present logarithmic shift scale. Thus at $D = qP$, where $D/L \asymp q^{-1}$, the fifth factorial moment is already below the second-order q^{-2} scale.

4.1 Fixed-endpoint singular-series averages and a reduced transfer hypothesis

For $r \geq 2$, define

$$A_r(D) = \sum_{0 < h_1 < \dots < h_{r-2} < D} \mathfrak{S}(\{0, h_1, \dots, h_{r-2}, D\}).$$

Lemma 4.1 (Iterated fixed-endpoint average). *For every fixed $r \geq 3$ and every $\varepsilon > 0$,*

$$A_r(D) = \mathfrak{S}(D) \frac{D^{r-2}}{(r-2)!} \left(1 + O_{r,\varepsilon}(D^{-1/2+\varepsilon})\right).$$

Proof. Feng–Wu’s Lemma 3.1 states that for a fixed finite set $\mathcal{D} \subset [0, D]$ and $H \leq D$,

$$\sum_{\substack{1 \leq h \leq H \\ h \notin \mathcal{D}}} \mathfrak{S}(\mathcal{D} \cup \{h\}) = \mathfrak{S}(\mathcal{D}) H \left(1 + O_{|\mathcal{D}|,\varepsilon} \left(\frac{D^\varepsilon}{H^{1/2}}\right)\right).$$

Start with $\mathcal{D} = \{0, D\}$ and iterate this identity a fixed $r - 2$ times with $H = D$. The exclusion $h \notin \mathcal{D}$ enforces distinct interior shifts, and every unordered $r - 2$ -tuple is counted exactly $(r - 2)!$ times by the ordered iteration. Since r is fixed, the finitely many relative errors combine into $1 + O_{r,\varepsilon}(D^{-1/2+\varepsilon})$, proving the claim. $\square$

Odlyzko–Rubinstein–Wolf obtained a fixed-endpoint asymptotic of the form

$$A_r(D) = \mathfrak{S}(D) \frac{D^{r-2}}{(r-2)!} + R_r(D), \quad R_r(D) \ll_r \frac{D^{r-2}}{\log \log D}.$$

The sharper estimate

$$R_r(D) \ll_{r,\varepsilon} D^{r-3+\varepsilon}$$

was announced by Goldston–Ledoan in the 2012 preliminary report; Feng–Wu note that no proof accompanied that announcement. The same sharper bound is stated again in Goldston–Ledoan’s later jumping-champion paper [8]. We do not rely on this sharper estimate here.

For the present second-order transition problem, the explicit Feng–Wu estimate, in the iterated fixed-endpoint form of Lemma 4.1, is sufficient and has the advantage of following directly from their published Lemma 3.1. In particular,

$$A_3(D) = \mathfrak{S}(D) D \left(1 + O_\varepsilon(D^{-1/2+\varepsilon})\right),$$

and

$$A_4(D) = \frac{1}{2} \mathfrak{S}(D) D^2 \left(1 + O_\varepsilon(D^{-1/2+\varepsilon})\right).$$

At the successive-primorial transition,

$$D = qP \asymp \frac{L}{q}, \quad q \sim \log \log X.$$

Therefore

$$D^{-1/2+\varepsilon} = o(q^{-A})$$

for every fixed $A > 0$, provided $0 < \varepsilon < 1/2$. Thus the arithmetic error in the fixed-endpoint singular-series average is far below the q^{-2} transition scale.

Define

$$M_D = \mathfrak{S}(D) \frac{X}{L^2}, \quad \lambda_D = \frac{D}{L}.$$

The exact Bonferroni inequality gives

$$B_D - T_3 + T_4 - T_5 \leq N(X, D) \leq B_D - T_3 + T_4.$$

By (4.4),

$$T_5(X, D) \ll M_D \lambda_D^3.$$

Hence at $D = qP$,

$$T_5(X, qP) = O(M_{qP} q^{-3}),$$

below the second-order scale.

This shows that a reduced sufficient input may be formulated as an averaged prime-correlation hypothesis rather than a full uniform Hardy–Littlewood conjecture.

Conjecture 4.2 (Second-order averaged transfer hypothesis). *Uniformly for*

$$L = Pq^2 \left(1 + \frac{\theta}{q}\right), \quad |\theta| \leq C,$$

with each fixed $C > 0$, assume for $D \in \{P, qP\}$ that

$$B_D = M_D \left(1 + o(q^{-2})\right).$$

At $D = P$, assume

$$T_3(X, P) = \frac{X}{L^3} A_3(P) + o(M_P q^{-2}).$$

At $D = qP$, assume

$$T_3(X, qP) = \frac{X}{L^3} A_3(qP) + o(M_{qP} q^{-2}),$$

and

$$T_4(X, qP) = \frac{X}{L^4} A_4(qP) + o(M_{qP} q^{-2}).$$

Theorem 4.3 (Conditional second-order transfer under the reduced hypothesis). *Assume Conjecture 4.2. Then, uniformly for*

$$L = Pq^2 \left(1 + \frac{\theta}{q}\right), \quad |\theta| = O(1),$$

one has

$$\frac{N(X, D)}{M_D} = 1 - \lambda_D + \frac{\lambda_D^2}{2} + o(q^{-2})$$

for $D \in \{P, qP\}$, and therefore

$$\log \frac{N(X, qP)}{N(X, P)} = \frac{\theta + 5/2}{q^2} + o(q^{-2}).$$

Consequently, for every fixed $\varepsilon > 0$, the logarithmic ratio is negative at

$$\theta = -\frac{5}{2} - \varepsilon$$

and positive at

$$\theta = -\frac{5}{2} + \varepsilon$$

for all sufficiently large q . Hence at least one pairwise sign change occurs between these two points. Moreover, any sequence of pairwise transition points $X = X_q$ lying in a bounded- θ window satisfies

$$\log X = P \left(q^2 - \frac{5}{2}q + o(q) \right).$$

Proof. The Feng–Wu averages give

$$\frac{X}{L^3} \frac{A_3(D)}{M_D} = \lambda_D + o(q^{-2})$$

for both $D = P$ and $D = qP$, and

$$\frac{X}{L^4} \frac{A_4(qP)}{M_{qP}} = \frac{\lambda_{qP}^2}{2} + o(q^{-2}).$$

At the smaller candidate,

$$\lambda_P \asymp q^{-2}.$$

The same upper-bound-sieve argument, now combined with Lemma 4.1 for $r = 4$, gives

$$T_4(X, P) \ll M_P \lambda_P^2 = O(M_P q^{-4}).$$

The hypotheses on B_D, T_3, T_4 , together with

$$T_5 = o(M_D q^{-2}),$$

therefore yield

$$N(X, D) = M_D \left(1 - \lambda_D + \frac{\lambda_D^2}{2} + o(q^{-2}) \right).$$

Since

$$\log \left(1 - \lambda + \frac{\lambda^2}{2} \right) = -\lambda + O(\lambda^3),$$

the logarithmic ratio is

$$\log \frac{\mathfrak{S}(qP)}{\mathfrak{S}(P)} - \frac{(q-1)P}{L} + o(q^{-2}).$$

Now

$$\log \frac{q-1}{q-2} = \frac{1}{q} + \frac{3}{2q^2} + O(q^{-3}),$$

and

$$\frac{(q-1)P}{L} = \frac{1}{q} - \frac{\theta+1}{q^2} + O(q^{-3}),$$

which gives the result. □

Corollary 4.4 (A transparent stronger sufficient hypothesis). *The preceding theorem follows in particular from any uniform Hardy–Littlewood hypothesis for prime tuples of sizes 2, 3, 4 with a fixed power-saving error*

$$\pi(X; \mathcal{H}) = \mathfrak{S}(\mathcal{H}) \frac{X}{L^{|\mathcal{H}|}} + O_{|\mathcal{H}|}(X^{1-\delta})$$

uniformly for the logarithmic shifts occurring here.

Remark 4.5. *The averaged transfer hypothesis is substantially weaker than the power-saving formulation. In particular, at the larger candidate $D = qP$, the triple main term is of size M_D/q , so second-order transition information requires only a relative $o(1/q)$ refinement of the averaged triple correlation. The quadruple main term is already of size M_D/q^2 , so only a relative $o(1)$ asymptotic is required there.*

5 Diagnostic of the remaining arithmetic obstruction

The conditional theorem identifies a sufficient second-order prime-correlation input. We now examine one concrete route from a controllable divisor proxy toward that input and isolate a genuinely prime-sensitive remainder; auxiliary local and Fourier/Type-I diagnostics are moved to Appendix A. The Vaughan analysis below is diagnostic: it exhibits a representative unresolved block rather than claiming a canonical or exhaustive reduction of every possible transfer strategy.

5.1 A pointwise divisor proxy at the transition scale

Let

$$\Lambda_R(m) = \sum_{\substack{r|m \\ r \leq R}} \mu(r) \log \frac{R}{r}, \quad G_R = \sum_{r \leq R} \frac{\mu(r)}{r} \log \frac{R}{r},$$

and normalize

$$\Lambda_R^*(m) = \frac{\Lambda_R(m)}{G_R}.$$

By the prime number theorem, $G_R = 1 + o(1)$, and hence it is bounded away from zero for sufficiently large R . Define

$$V_{1,R}(n) = \sum_{1 \leq h < D} \Lambda_R^*(n+h)$$

and

$$V_{2,R}(n) = \sum_{1 \leq h_1 < h_2 < D} \Lambda_R^*(n+h_1) \Lambda_R^*(n+h_2).$$

Theorem 5.1 (Pointwise divisor-proxy moments). *Uniformly in n , as $R \rightarrow \infty$ with $R \leq D$,*

$$V_{1,R}(n) = D + O(R)$$

and

$$V_{2,R}(n) = \frac{D^2}{2} + O\left(DR(\log R)^2 + R^2\right).$$

The proof is elementary but somewhat longer than the role of the result in the main argument; it is given in Appendix B. The key uniform residue count is

$$\#\{1 \leq h_1 < h_2 < D : r \mid n+h_1, s \mid n+h_2\} = \frac{D^2}{2rs} + O\left(\frac{D}{r} + \frac{D}{s} + 1\right).$$

Corollary 5.2 (Transition-scale proxy accuracy). *Let*

$$D = qP = q^\#, \quad L \asymp Pq^2, \quad R = \frac{D}{q^4}.$$

Then, uniformly in n ,

$$V_{1,R}(n) = D + O(Dq^{-4}), \quad V_{2,R}(n) = \frac{D^2}{2} + O(D^2q^{-2}),$$

and consequently

$$V_{1,R}(n) - \frac{V_{2,R}(n)}{L} = D - \frac{D^2}{2L} + O(Dq^{-3}).$$

Proof. Theorem 5.1 and $\log R \sim q$ give

$$DR(\log R)^2 \ll D^2q^{-2}, \quad R^2 \ll D^2q^{-8}.$$

Since $D/L \asymp q^{-1}$, division of the second-moment error by L gives $O(Dq^{-3})$. □

5.2 The prime-versus-proxy remainder

Define

$$E_D(n) = \Lambda(n)\Lambda(n+D),$$

$$W_1(n) = \sum_{1 \leq h < D} \Lambda(n+h),$$

and

$$W_2(n) = \sum_{1 \leq h_1 < h_2 < D} \Lambda(n+h_1)\Lambda(n+h_2).$$

Set

$$\mathcal{C}(X, D) = \sum_{n \leq X-D} E_D(n) \left(W_1(n) - \frac{W_2(n)}{L} \right),$$

and

$$\mathcal{C}_R(X, D) = \sum_{n \leq X-D} E_D(n) \left(V_{1,R}(n) - \frac{V_{2,R}(n)}{L} \right).$$

Finally,

$$\mathcal{P}_R(X, D) = \mathcal{C}(X, D) - \mathcal{C}_R(X, D).$$

Let

$$B_D^\Lambda(X) = \sum_{n \leq X-D} \Lambda(n)\Lambda(n+D).$$

The preceding corollary gives

$$\mathcal{C}_R(X, D) = B_D^\Lambda(X) \left(D - \frac{D^2}{2L} \right) + O \left(B_D^\Lambda(X) \frac{D}{q^3} \right).$$

Thus, once the weighted endpoint-pair asymptotic $B_D^\Lambda = \mathfrak{S}(D)X(1 + o(q^{-1}))$ is supplied, the only additional second-order prime-correlation input at $D = qP$ in this reduction is

$$\mathcal{P}_R(X, D) = o \left(\mathfrak{S}(D) \frac{XL}{q^2} \right).$$

Lemma 5.3 (Removing von Mangoldt weights). *At $D = qP$, assume the standard fixed-dimensional upper-bound sieve estimates for the relevant three- and four-point patterns. Then*

$$\mathcal{C}(X, D) = L^3(T_3(X, D) - T_4(X, D)) + o\left(\mathfrak{S}(D)\frac{XL}{q^2}\right).$$

If, in addition,

$$B_D^\Lambda(X) = \mathfrak{S}(D)X(1 + o(q^{-1}))$$

and

$$\mathcal{P}_R(X, D) = o\left(\mathfrak{S}(D)\frac{XL}{q^2}\right),$$

then

$$\frac{T_3(X, D) - T_4(X, D)}{M_D} = \lambda_D - \frac{\lambda_D^2}{2} + o(q^{-2}).$$

Proof. See Appendix C for the first assertion. For the second, Corollary 5.2 and the weighted pair asymptotic give

$$\mathcal{C}_R(X, D) = \mathfrak{S}(D)X\left(D - \frac{D^2}{2L}\right) + o\left(\mathfrak{S}(D)\frac{XL}{q^2}\right),$$

because $D/L \asymp q^{-1}$. Adding the assumed bound for $\mathcal{P}_R = \mathcal{C} - \mathcal{C}_R$ and then using the first assertion yields

$$L^3(T_3 - T_4) = \mathfrak{S}(D)X\left(D - \frac{D^2}{2L}\right) + o\left(\mathfrak{S}(D)\frac{XL}{q^2}\right).$$

Division by $L^3M_D = \mathfrak{S}(D)XL$ gives the claimed normalized formula. $\square$

5.3 Signed reassembly and the affine short-shift obstruction

The remaining prime-versus-proxy term may be reorganized by Vaughan type decompositions. It is important, however, not to mistake this reorganization for an unconditional parity-breaking estimate.

Let

$$\beta_q(k) = \sum_{\substack{c|k \\ c > q}} \Lambda(c)$$

and

$$\Lambda_{>q}(m) = \Lambda(m)\mathbf{1}_{m>q}.$$

Then

$$\beta_q = 1 * \Lambda_{>q}$$

in Dirichlet-convolution notation, and therefore

$$\mu * \beta_q = \Lambda_{>q}. \tag{V.1}$$

Proposition 5.4 (Exact signed reassembly). *Let Δ_m be any finitely supported sequence. Then*

$$\sum_{b,k} \mu(b)\beta_q(k)\Delta_{bk} = \sum_m \Lambda_{>q}(m)\Delta_m. \tag{V.2}$$

In particular, on ranges $m \asymp X$ with $X \gg q$, the right hand side is

$$\sum_m \Lambda(m)\Delta_m.$$

Proof. See Appendix C. □

Remark 5.5. *Thus retaining the Möbius signs in Vaughan's identity does not by itself produce a weaker global problem. After all dyadic blocks are reassembled, the original von Mangoldt detector returns exactly. Any gain must therefore come from a genuinely new estimate inside the separated bilinear blocks, not from the decomposition alone; compare the classical sieve and Vaughan frameworks in [18, 19].*

For one such block, a representative true-prime contribution has the form

$$\mathcal{T}(B, K) = \sum_{k \sim K} \beta_q(k) \sum_{b \sim B} \mu(b) \sum_{1 \leq h < D} \Lambda(kb - h) \Lambda(kb - h + D), \quad BK \asymp X. \quad (\text{V.3})$$

Write

$$h = a + ks, \quad 0 \leq a < k,$$

and set

$$t = b - s.$$

Then

$$kb - h = kt - a, \quad \mu(b) = \mu(t + s).$$

After inserting smooth cutoffs to absorb the dyadic boundary conditions, (V.3) is reorganized into sums of the schematic form

$$\sum_{k \sim K} \beta_q(k) \sum_{a \bmod k} \sum_{s \in I_{k,a}} \sum_t \mu(t + s) \Lambda(kt - a) \Lambda(kt - a + D) W_{k,a,s}(t), \quad (\text{V.4})$$

where $W_{k,a,s}$ is a smooth weight inherited from the original dyadic block and

$$|I_{k,a}| = \frac{D}{k} + O(1). \quad (\text{V.5})$$

Since

$$\beta_q(k) = 0 \quad (k \leq q),$$

every nonzero block in (V.4) has $k > q$. Consequently

$$|I_{k,a}| \leq \frac{D}{q} + O(1) = P + O(1). \quad (\text{V.6})$$

At the successive-primorial transition,

$$L \asymp Pq^2, \quad q \sim \log \log X,$$

and hence

$$P \asymp \frac{\log X}{(\log \log X)^2}. \quad (\text{V.7})$$

Equation (V.4) therefore exposes a genuine short-shift correlation between one Möbius factor and two prime factors. It is natural to describe this as an *affine Hardy–Littlewood–Chowla type* correlation.

There are, however, two independent reasons why currently available averaged Hardy–Littlewood–Chowla theorems do not close (V.4).

First, the theorem of Lichtman–Teräväinen [21] treats correlations of the shape

$$\sum_{n \leq X} \mu(n + h_1) \cdots \mu(n + h_r) \Lambda(n + a_1) \cdots \Lambda(n + a_\ell)$$

with the other shifts fixed, averaged over one additive shift $h_1 \leq H$. Their range is

$$H \geq (\log X)^{\ell+\varepsilon}.$$

With two von Mangoldt factors this means

$$H \geq (\log X)^{2+\varepsilon}.$$

Even at the level of shift length alone, our maximal range

$$H \leq P \asymp \frac{\log X}{(\log \log X)^2}$$

is substantially shorter.

Second, (V.4) is not literally of the fixed-shift form covered by that theorem. The prime arguments are

$$kt - a, \quad kt - a + D,$$

with the coefficient k and the separation $D = q^\#$ both varying with the main parameter. Thus an application would additionally require uniformity in growing affine coefficients and arithmetic progressions that is not supplied by the cited averaged HLC theorem.

Recent higher-uniformity results for Möbius and von Mangoldt functions in almost all short intervals, such as [22], operate in structurally different regimes; in the general von Mangoldt setting they include scales such as $H \geq X^{1/3+\varepsilon}$, far above the logarithmic range here. They therefore do not directly imply the estimate required for (V.4).

Accordingly, the displayed Vaughan block leads to the following diagnostic conclusion.

Remark 5.6 (Diagnostic short-shift obstruction in the displayed Vaughan block). *For the representative bilinear block displayed in (V.3)–(V.4), the shift interval satisfies*

$$H \leq P \asymp \frac{\log X}{(\log \log X)^2}.$$

No theorem cited in the present paper supplies the uniform cancellation in the growing parameters k, D needed to control this displayed block at the second-order transition scale. This is evidence for the specific obstruction encountered by the present route, not a theorem that every possible transfer must reduce to (V.4).

Remark 5.7 (Diagnostic, not equivalence). *The preceding diagnostic should not be read as an equivalence between the successive-primorial transition problem and the published Hardy–Littlewood–Chowla theorem. The HLC terminology identifies the type of parity-breaking correlation that remains. Existing HLC results provide a useful benchmark for the shift scale, but they do not directly apply to the growing affine forms in (V.4).*

Remark 5.8 (Why outer averaging does not automatically solve the problem). *One might hope that the additional sums over k and a in (V.4) compensate for the short s -interval. However, if these outer variables are recombined without a new estimate, the convolution identity*

$$\mu * \beta_q = \Lambda_{>q}$$

reconstructs the original prime detector. Thus the outer averaging is potentially useful only if one proves cancellation before this reassembly; it is not, by itself, an additional independent averaging parameter.

6 Discussion and conclusion

The unconditional result of this paper is an all-orders local theory for the crossover root in the explicitly defined ORW exponential-integral approximation. The endpoint integral has a uniform generalized-Laguerre expansion with differentiated remainder control, and the logarithm of that expansion admits a graded coefficient structure adapted to a growing endpoint. This is what makes rigorous fixed-order reversion possible and yields the explicit Lagrange–Bürmann generator for the root coefficients.

For successive primorials, the full integral root separates naturally into two scales. ORW’s classical leading value

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}$$

has the algebraic expansion

$$\frac{L_0}{P} = q^2 - \frac{5}{2}q + \frac{17}{12} + \cdots,$$

whereas solving the full endpoint integral produces the absolute displacement

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + \cdots.$$

The general-pair theorem shows that the constants 1 and 3 are endpoint-universal under $D_1/L_0 \rightarrow 0$; specific primorial arithmetic first enters at the next polynomial coefficient.

The relation to actual consecutive-prime gaps is intentionally separate. Under the reduced averaged prime-correlation hypothesis of Section 4, the second-order coefficient transfers pairwise, but this remains conditional and does not establish global jumping-champion dominance. Section 5 shows what one concrete unconditional transfer route still encounters: after pointwise divisor-proxy control and exact signed reassembly, a representative short affine Möbius–two-prime block lies outside the uniformity supplied by the results cited here.

The scope of the novelty claim is correspondingly narrow. The paper does not claim the primorial transition mechanism, the ORW leading scale L_0 , the coefficient $-5/2$, generalized Laguerre polynomials, or formal inversion in isolation. The claimed contribution is the uniform Laguerre endpoint organization in the growing-parameter ORW regime, the graded all-orders inversion of the full integral root, the universal first endpoint corrections, and the explicit coefficient/degree hierarchy. Because Goldston and Ledoan announced a sequel in the same small-normalized-gap regime and later related work contains sharper singular-series information, no absolute historical-priority claim is made beyond formulas that can be compared directly in the published and publicly accessible literature.

A Auxiliary diagnostics for the arithmetic reduction

The material in this appendix supports the interpretation of the arithmetic obstruction but is not needed for the unconditional ORW root theorem. It is separated from the main text to keep the logical hierarchy explicit.

A.1 Primorial pre-sieving and a local diagnostic

Set $D = qP = q^\#$. If $(n, D) = 1$, then exactly $\varphi(D) - 1$ shifts $1 \leq h < D$ satisfy $(n + h, D) = 1$. Thus all possible interior primes lie in a completely pre-sieved set modulo D .

There is also a useful local translation identity. Let $H = \{0, D\}$, and define the formal local factor with four-form normalization by

$$\sigma_p(0, D, r, r + D) = \left(1 - \frac{\nu_p(\{0, D, r, r + D\})}{p}\right) \left(1 - \frac{1}{p}\right)^{-4}.$$

Then

$$\frac{1}{p} \sum_{r \bmod p} \sigma_p(0, D, r, r + D) = \sigma_p(0, D)^2.$$

Indeed, if $\nu = |H \bmod p|$, then

$$\frac{1}{p} \sum_{r \bmod p} |H \cap (H + r)| = \frac{\nu^2}{p},$$

so the averaged unnormalized local density factorizes. We use this identity only as a local diagnostic; it is not needed for the divisor-proxy theorem below, and the four-form normalization is retained even in degenerate residue classes.

A.2 Why a very short divisor proxy cannot detect primality

For

$$\Lambda_R(m) = \sum_{\substack{r|m \\ r \leq R}} \mu(r) \log \frac{R}{r},$$

if $R \leq q$ and $(m, D) = 1$, every divisor $1 < r \leq R$ is excluded, and hence

$$\Lambda_R(m) = \log R.$$

Thus a proxy truncated at or below q is constant on the primorial-admissible positions: it cannot distinguish a prime from a rough composite. This is a concrete parity diagnostic and motivates the longer normalized proxy used next. The higher-correlation theory of such truncated divisor sums was developed by Goldston and Yıldırım [13, 14, 16].

A.3 Fourier and Type-I sieve interpretations

Define

$$A_D(\alpha) = \sum_{n \leq X-D} \Lambda(n) \Lambda(n + D) e^{2\pi i n \alpha},$$

$$S(\alpha) = \sum_{m \leq X} \Lambda(m) e^{2\pi i m \alpha}, \quad \mathcal{D}_D(\alpha) = \sum_{1 \leq h < D} e^{2\pi i h \alpha}.$$

Then the weighted triple correlation is

$$\sum_n \Lambda(n) \Lambda(n+D) \sum_{h < D} \Lambda(n+h) = \int_0^1 \mathcal{D}_D(\alpha) A_D(\alpha) S(-\alpha) d\alpha.$$

Replacing S by the Fourier transform of the normalized divisor proxy shows that the unresolved term requires nonzero-frequency pseudorandomness of the endpoint-pair sequence $n \mapsto \Lambda(n) \Lambda(n+D)$. A pair Hardy–Littlewood asymptotic controls its zero-frequency mass but does not by itself provide this information.

There is a complementary sieve formulation. Set

$$a_m^{(D)} = \sum_{1 \leq h < D} \Lambda(m-h) \Lambda(m-h+D).$$

Then

$$\sum_m a_m^{(D)} = (D-1) B_D^\Lambda.$$

For a modulus c , reordering by the endpoint variable gives

$$\sum_{c|m} a_m^{(D)} = \frac{1}{c} \sum_m a_m^{(D)} + O(B_D^\Lambda).$$

Thus the natural Type-I density is $g(c) = 1/c$. Summing this relation over $c \leq R$ gives a relative Type-I discrepancy $O(R/D)$. At logarithmic levels such as $R = D/q^3$, this is already $O(q^{-3})$, below the desired transition scale. The remaining difficulty is therefore of the parity-breaking/bilinear type emphasized by asymptotic-sieve theory [18].

B Technical proofs for the divisor-proxy reduction

For completeness we give the elementary residue-count proof underlying Theorem 5.1. Put

$$w_r = \frac{\mu(r) \log(R/r)}{G_R}, \quad \Lambda_R^*(m) = \sum_{\substack{r|m \\ r \leq R}} w_r.$$

Then

$$\sum_{r \leq R} \frac{w_r}{r} = 1, \quad \sum_{r \leq R} |w_r| \ll R, \quad \sum_{r \leq R} \frac{|w_r|}{r} \ll (\log R)^2.$$

Lemma B.1 (One-residue count). *Uniformly in n and $1 \leq r \leq D$,*

$$\#\{1 \leq h < D : r \mid n+h\} = \frac{D}{r} + O(1).$$

Consequently $V_{1,R}(n) = D + O(R)$.

Proof. The first assertion is the elementary count of one residue class in an interval of length $D + O(1)$. Hence

$$V_{1,R}(n) = \sum_{r \leq R} w_r \left(\frac{D}{r} + O(1) \right) = D + O \left(\sum_{r \leq R} |w_r| \right) = D + O(R).$$

□

Lemma B.2 (Ordered two-residue count). *Uniformly in n and $1 \leq r, s \leq D$,*

$$C_{r,s}(n) := \#\{1 \leq h_1 < h_2 < D : r \mid n + h_1, s \mid n + h_2\} = \frac{D^2}{2rs} + O\left(\frac{D}{r} + \frac{D}{s} + 1\right).$$

Proof. Let $\mathcal{A}_r = \{1 \leq h < D : h \equiv -n \pmod{r}\}$. Then $|\mathcal{A}_r| = D/r + O(1)$. For fixed $h_1 \in \mathcal{A}_r$, the number of $h_2 \in (h_1, D)$ in the required residue class modulo s is $(D - h_1)/s + O(1)$. Therefore

$$C_{r,s}(n) = \frac{1}{s} \left(D|\mathcal{A}_r| - \sum_{h \in \mathcal{A}_r} h \right) + O(|\mathcal{A}_r|).$$

An arithmetic-progression sum gives

$$\sum_{h \in \mathcal{A}_r} h = \frac{D^2}{2r} + O(D + r),$$

which proves the claim because $r \leq D$. □

Proof of Theorem 5.1. Expanding the two divisor sums gives

$$V_{2,R}(n) = \sum_{r,s \leq R} w_r w_s C_{r,s}(n).$$

The main term is

$$\frac{D^2}{2} \left(\sum_{r \leq R} \frac{w_r}{r} \right)^2 = \frac{D^2}{2}.$$

The total error is

$$\ll D \left(\sum_{r \leq R} |w_r| \right) \left(\sum_{s \leq R} \frac{|w_s|}{s} \right) + \left(\sum_{r \leq R} |w_r| \right)^2,$$

which is $O(DR(\log R)^2 + R^2)$. □

C Weight removal and signed reassembly

Proof of the weight-removal lemma. Choose a fixed large B and split the n -sum at

$$Y = \frac{X}{L^B}.$$

The contribution of $n \leq Y$ is $o(\mathfrak{S}(D)XL/q^2)$ by a fixed-dimensional upper-bound sieve, since $D = O(L)$ and L^{-B} dominates every fixed power of q^{-1} .

For $n > Y$ and every shift $0 \leq h \leq D$,

$$\log(n + h) = L + O(\log L)$$

uniformly. Hence every product of three prime logarithms equals

$$L^3 \left(1 + O\left(\frac{\log L}{L}\right) \right),$$

and every product of four prime logarithms, after division by L , has the same form. Replacing Λ by the prime-supported Chebyshev weight changes the sums only when one of the variables is a proper prime power. A trivial count of proper prime powers, together with $D = O(L)$, bounds their total contribution by $O(X^{1/2}D^2L^{O(1)})$, negligible at the target scale. Since $\log L/L = o(q^{-A})$ for every fixed $A > 0$, the claimed relation follows. □

Proof of the exact signed-reassembly proposition. Using $\beta_q = 1 * \Lambda_{>q}$, Dirichlet convolution gives

$$\mu * \beta_q = \Lambda_{>q}.$$

Grouping the left-hand side of (V.2) by $m = bk$ therefore yields

$$\sum_{b,k} \mu(b) \beta_q(k) \Delta_{bk} = \sum_m \left(\sum_{bk=m} \mu(b) \beta_q(k) \right) \Delta_m = \sum_m \Lambda_{>q}(m) \Delta_m.$$

□

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

No research data were generated or analyzed for this theoretical study. The ancillary script `numerical_check.py`, supplied with the submission, reproduces the numerical consistency table; none of the proofs depends on the numerical computation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to support literature exploration, mathematical cross-checking and proof auditing, ancillary numerical-code preparation and checking, and manuscript organization and language refinement. The author reviewed and edited the resulting material, verified the mathematical claims and cited sources used in the final manuscript, and takes full responsibility for the content of the publication.

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